

Highly tunable efficient second-harmonic generation in a lithium niobate nanophotonic waveguide

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Highly tunable coherent light generation is crucial for many important photonic applications. Second-harmonic generation (SHG) is a dominant approach for this purpose, which, however, exhibits a trade-off between conversion efficiency and wavelength tunability in a conventional nonlinear platform. Recent development of the integrated lithium niobate (LN) technology makes it possible to achieve a large wavelength tuning while maintaining a high conversion efficiency. Here we report on-chip SHG that simultaneously achieves a large tunability and a high conversion efficiency inside a single device. We utilize the unique strong thermo-optic birefringence of LN to achieve flexible temperature tuning of type-I intermodal phase matching. We experimentally demonstrate spectral tuning with a tuning slope of 0.84 nm/K for a telecom-band pump, and a nonlinear conversion efficiency of 4.7% W⁻¹, in an LN nanophotonic waveguide only 8 mm long. Our device shows great promise for efficient on-chip wavelength conversion to produce highly tunable coherent visible light for broad applications, while taking advantage of the mature and cost-effective telecom laser technology. © 2018 Optical Society of America under the terms of the [OSA Open Access Publishing Agreement](#)

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1. INTRODUCTION

Since their invention in the 1960s [1,2], lasers have been the backbone of modern optics, playing fundamental roles in optical sciences and technologies. For a coherent light source, wavelength tunability is one of the most important specifications, crucially underlying many applications including optical communications [3], spectroscopy [4,5], frequency metrology [6], sensing [7], to name some. However, lasing wavebands are naturally limited by gain media. Nonlinear optical parametric processes, such as second-harmonic generation (SHG), sum-frequency generation (SFG), and difference-frequency generation (DFG), with the flexible engineering of the phase-matching condition, are probably the most prominent approaches to achieve tunable coherent radiation at optical frequencies that can hardly be obtained by lasers directly [8–10].

Lithium niobate (LN), with outstanding nonlinear and linear optical properties, is widely employed for this application, where SFG/DFG has been extensively studied over the past decades, particularly in periodically poled lithium niobate (PPLN) waveguides [11,12]. In general, a type-0 configuration is employed to achieve a high conversion efficiency, and temperature tuning is a common technique to vary the operation wavelength of a PPLN waveguide. However, the pump wavelength tunability of the

type-0 SHG is fairly limited [11–13], mainly due to the relatively small wavelength dependence of the thermo-optic coefficient (although DFG can exhibit a large wavelength tunability with a third wave involved). In fact, LN exhibits a remarkable thermo-optic birefringence [14–16], significantly greater than most other optical media [17]. This characteristic can be utilized to greatly increase the wavelength tunability of SHG in PPLN by employing a type-I configuration [11,12,18], which, however, inevitably seriously sacrifices the conversion efficiency due to the significantly weaker nonlinearity compared with a type-0 process.

Over the past decade, a variety of integrated material platforms with $\chi^{(2)}$ nonlinearity have been developed for efficient nonlinear optical parametric processes [19–25], where the tight confinement of optical modes is able to significantly enhance nonlinear optical interactions. Recent advances in the integrated LN platform have greatly inspired study of nonlinear optics in LN nanophotonic structures [26–35], showing great potential for nonlinear wavelength conversion with even higher efficiencies compared with PPLN and other integrated platforms. This provides an opportunity to achieve a large wavelength tunability by using the type-I configuration while maintaining a high conversion efficiency at the same time.

Here we demonstrate highly tunable efficient on-chip SHG in an LN nanophotonic waveguide. We achieve SHG through type-I intermodal phase matching between orthogonal polarizations, and by utilizing the strong thermo-optic birefringence of LN, we demonstrate temperature tuning of the SHG wavelength, with a measured tuning slope of 0.84 nm/K for a telecom pump, almost 1 order of magnitude higher than that of type-0 SHG in LN [11,12,36]. Meanwhile, our device is designed to exhibit a large mode overlap, resulting in a theoretical normalized SHG efficiency of $22.2\% \text{ W}^{-1} \text{ cm}^{-2}$, which enables us to experimentally demonstrate a conversion efficiency of $4.7\% \text{ W}^{-1}$ in a waveguide only 8 mm long. Our device is of great promise for efficient on-chip wavelength conversion to produce highly tunable coherent visible light, which is essential for various integrated photonic applications such as particle and chemical sensing in aqueous environments [37–39], while taking advantage of the mature telecom laser technology.

2. WAVEGUIDE DESIGN

LN exhibits a significant thermo-optic birefringence, with a value of $|\frac{dn_e}{dT} - \frac{dn_o}{dT}| \sim 4 \times 10^{-5} \text{ K}^{-1}$ at room temperature [14–16], where $\frac{dn_e}{dT}$ and $\frac{dn_o}{dT}$ are the thermo-optic coefficients for the extraordinary and ordinary light, respectively. As a result, if SHG occurs in an LN waveguide between optical waves with orthogonal polarizations, a temperature change of the device would result in a considerable variation of the material birefringence, which in turn shifts the phase-matched wavelength of the SHG process significantly. In particular, we can maximize this effect by using a Z-cut LN waveguide, which supports ordinarily and extraordinarily polarized optical modes with high polarization purity [see Figs. 1(a) and 1(b)] [15].

We design the geometry of the Z-cut LN waveguide [see Fig. 1(a)] such that the fundamental quasi-transverse-electric mode ($\text{TE}_{0,\text{tele}}$) in the telecom band is phase matched with the third-order quasi-transverse-magnetic mode ($\text{TM}_{2,\text{vis}}$) in the visible. Figures 1(c) and 1(d) show the effective refractive indices of the two modes, simulated by the finite-element method (FEM), which gives a phase-matched pump wavelength of $\lambda_{\text{PM}} = 1540 \text{ nm}$ at room temperature of 20°C . Of particular interest is that LN exhibits a significant thermo-optic effect for extraordinary light ($\frac{dn_{e,\text{vis}}}{dT} \approx 4 \times 10^{-5} \text{ K}^{-1}$), while it is negligible for ordinary light ($\frac{dn_{o,\text{tele}}}{dT} \approx 0$) around room temperature [14]. As a result, when the device temperature increases, the effective refractive index of the $\text{TE}_{0,\text{tele}}$ mode remains nearly intact, while that of the $\text{TM}_{2,\text{vis}}$ mode increases considerably. Consequently, the phase-matched wavelength moves dramatically towards longer wavelengths. Figure 1(e) shows an example, where λ_{PM} shifts to 1574 nm at a temperature of 70°C . Detailed analysis shows that the phase-matched wavelength depends almost linearly on the device temperature, as shown clearly in Fig. 1(f), with a significant tuning slope of 0.69 nm/K .

Phase matching of the two modes indicates potentially efficient SHG in the designed waveguide. For a lossless waveguide without pump depletion, the SHG efficiency is given by the following expression [8,41]:

$$\Gamma \equiv \frac{P_2}{P_1^2} = \eta L^2 \left[\frac{\sin(\Delta L/2)}{\Delta L/2} \right]^2, \quad (1)$$

where P_1 and P_2 are the optical powers input at the fundamental wavelength λ and produced at the second harmonic, respectively. L is the waveguide length and $\Delta \equiv \frac{4\pi}{\lambda} (n_2 - n_1)$ represents the

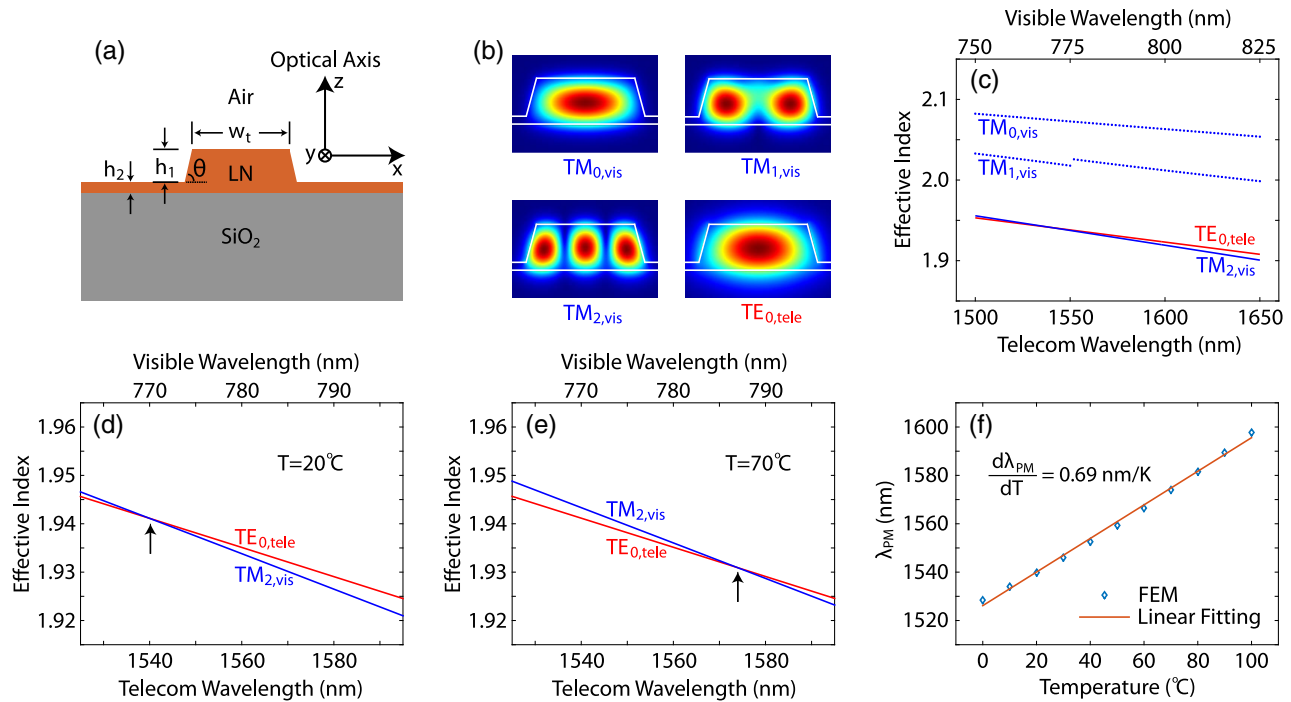


Fig. 1. (a) Schematic of our Z-cut LN waveguide. FEM simulations of (b) mode profiles, and (c) effective indices as functions of wavelength, of $\text{TE}_{0,\text{tele}}$ in the telecom and $\text{TM}_{j,\text{vis}}$ ($j = 0, 1, 2$) in the visible, where $w_t = 1200 \text{ nm}$, $h_1 = 460 \text{ nm}$, $h_2 = 100 \text{ nm}$, and $\theta = 75^\circ$, at 20°C . Discontinuity in the curve of $\text{TM}_{1,\text{vis}}$ is due to its coupling with $\text{TE}_{2,\text{vis}}$ (not shown). Zoom-in of the wavelength-dependent effective indices of $\text{TE}_{0,\text{tele}}$ and $\text{TM}_{2,\text{vis}}$ at (d) 20°C , and (e) 70°C , with black arrows indicating phase matching; (f) simulated phase-matched pump wavelength λ_{PM} as a function of temperature. In (c)–(f), the FEM simulations take into account the temperature and wavelength dependence of the material refractive indices, for both ordinary and extraordinary light [14,40].

phase mismatch, where n_1 and n_2 are the effective refractive indices of the $\text{TE}_{0,\text{tele}}$ mode at the fundamental wavelength and the $\text{TM}_{2,\text{vis}}$ mode at the second harmonic, respectively. When the phase-matching condition is satisfied ($\Delta = 0$), Eq. (1) shows the maximum SHG efficiency $\Gamma_0 = \eta L^2$ that depends on the normalized conversion efficiency given as

$$\eta = \frac{8\pi^2}{\epsilon_0 c n_1^2 n_2 \lambda^2} \frac{\zeta^2 d_{\text{eff}}^2}{A_{\text{eff}}}, \quad (2)$$

where ϵ_0 and c are the permittivity and light speed in vacuum, respectively, and d_{eff} is the effective nonlinear susceptibility. In Eq. (2), $A_{\text{eff}} \equiv (A_1^2 A_2)^{1/3}$ is the effective mode area, where $A_i = \frac{(\int_{\text{all}} |\vec{E}_i|^2 dx dz)^3}{|\int_{\chi^{(2)}} |\vec{E}_i|^2 \vec{E}_i dx dz|}$, ($i = 1, 2$), and ζ represents the spatial mode overlap factor between the fundamental and second-harmonic modes, given as

$$\zeta = \frac{\int_{\chi^{(2)}} (E_{1x}^*)^2 E_{2y} dx dz}{\left| \int_{\chi^{(2)}} |\vec{E}_1|^2 \vec{E}_1 dx dz \right|^{2/3} \left| \int_{\chi^{(2)}} |\vec{E}_2|^2 \vec{E}_2 dx dz \right|^{1/3}}, \quad (3)$$

where $\int_{\chi^{(2)}}$ and $\int_{\text{all}}$ denote two-dimensional integration over the LN material and all space, respectively, E_{1x} is the x component of

$\vec{E}_1(x, z)$, the electric field of the fundamental mode $\text{TE}_{0,\text{tele}}$, and E_{2y} is the y component of $\vec{E}_2(x, z)$, the electric field of the second-harmonic mode $\text{TM}_{2,\text{vis}}$.

Equations (1)–(3) show that the SHG efficiency depends essentially on the spatial mode overlap ζ , the effective mode area A_{eff} , and the effective nonlinear susceptibility d_{eff} . Numerical simulation shows that our waveguide exhibits a small $A_{\text{eff}} = 1.46 \mu\text{m}^2$. In particular, our designed waveguide exhibits a large spatial mode overlap, with $\zeta = 0.32$. As a result, the waveguide exhibits a normalized conversion efficiency as high as $\eta = 22.2\% \text{ W}^{-1} \text{ cm}^{-2}$. This value is comparable to that of type-0 SHG in typical PPLN [36,42] and LN nanophotonic waveguides [31] that utilize the maximum component of the $\chi^{(2)}$ nonlinearity ($d_{\text{eff}} = d_{33} = 27 \text{ pm/V}$), although a type-I configuration is employed here ($d_{\text{eff}} = d_{31} = 4.3 \text{ pm/V}$ [43]). In contrast to those type-0 devices, our waveguide is expected to exhibit a significantly larger thermal tuning slope, as we will experimentally demonstrate in the following.

3. EXPERIMENTAL RESULTS

To confirm our simulation results, we fabricated waveguides on a Z-cut LN-on-insulator wafer [see Fig. 2(b)], where the LN thin

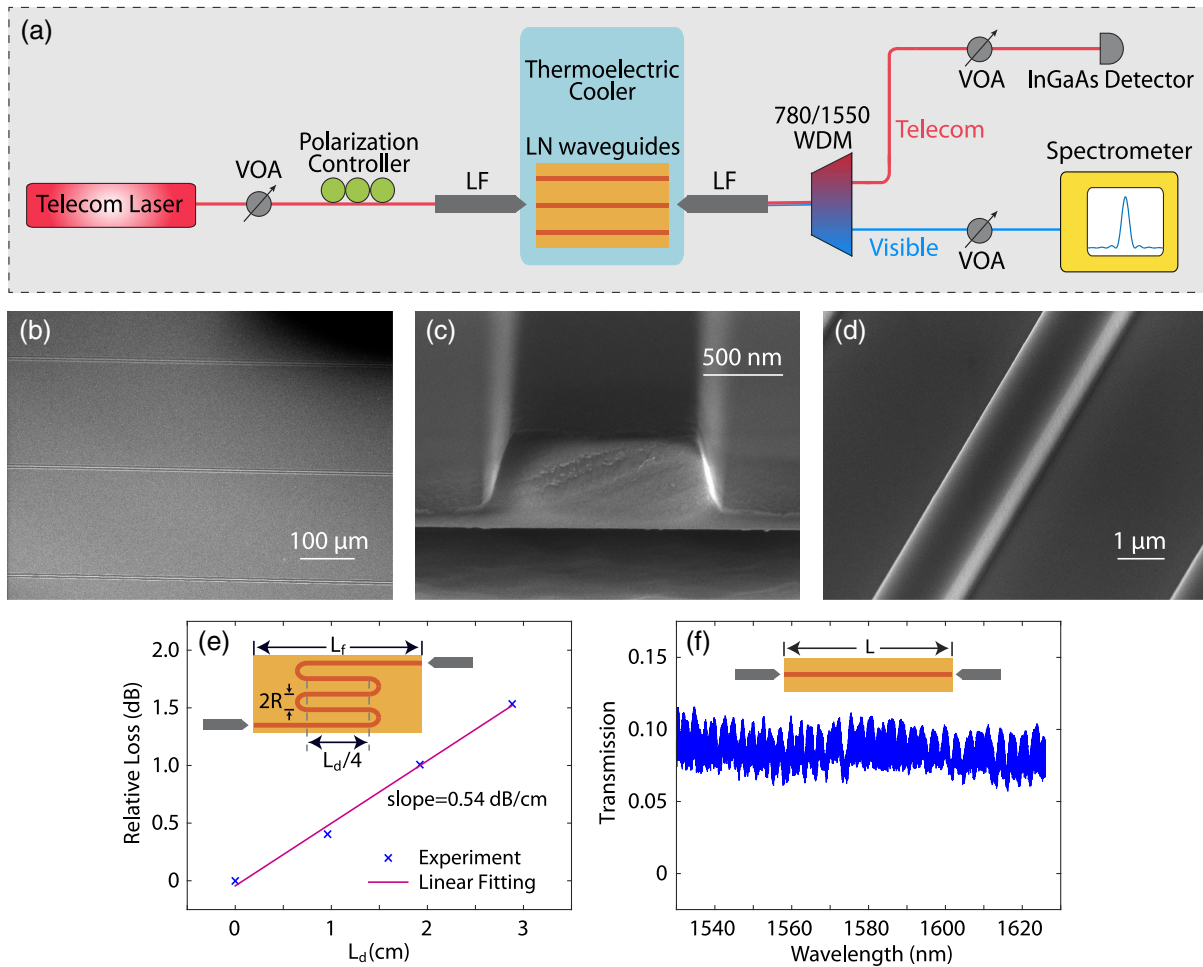


Fig. 2. (a) Experimental setup for device characterization and SHG measurement. Scanning electron microscope pictures showing the waveguide (b) top view, (c) facet, and (d) sidewall. (e) Fiber-to-fiber loss as a function of the differential length L_d , relative to that of $L_d = 0$, for waveguides schematically illustrated in the inset, where L_f and R are kept as 8 mm and 100 μm , respectively. (f) Telecom-band transmission spectrum of the TE polarization for a straight waveguide with a length of $L \simeq 8 \text{ mm}$, whose schematic is shown in the inset. VOA, variable optical attenuator; LF, lensed fiber; WDM, wavelength division multiplexer.

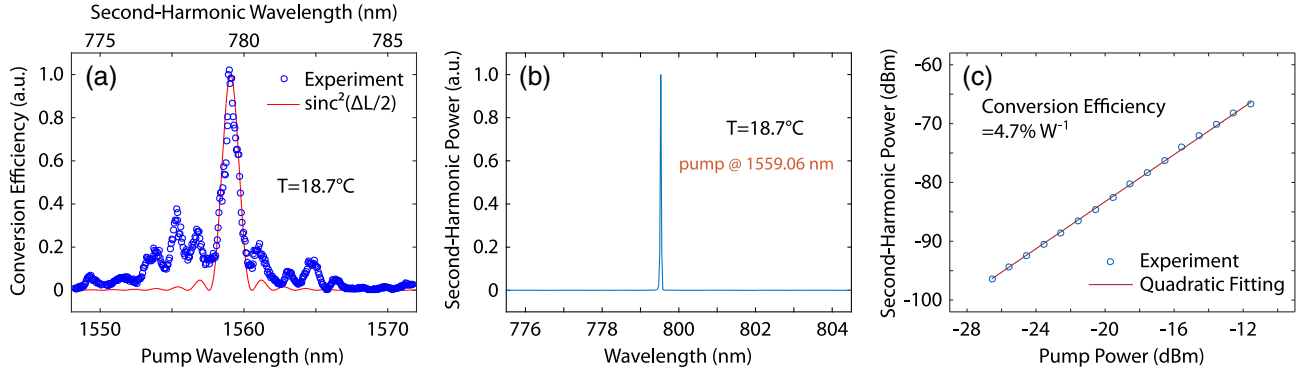


Fig. 3. SHG from a straight LN nanophotonic waveguide with a length of 8 mm. (a) Conversion efficiency spectrum at $T = 18.7^\circ\text{C}$, with the center wavelength of the sinc^2 function aligned to the measured peak; (b) SHG spectrum at a fixed pump wavelength of 1559.06 nm at $T = 18.7^\circ\text{C}$; (c) second-harmonic power as a function of pump power, with experimental data compared with a quadratic fitting, exhibiting a conversion efficiency of $4.7\% \text{ W}^{-1}$.

film has a thickness of ~ 560 nm, sitting on $2\text{-}\mu\text{m}$ -thick buried oxide. Figure 2(c) shows the cross section of a fabricated waveguide whose geometry is very close to our design [see Fig. 1(a)].

In particular, as presented in Fig. 2(d), the waveguide sidewall is very smooth, implying a low propagation loss. In order to quantify the propagation and coupling losses, we fabricated waveguides with the same cross section but different lengths, as schematically shown in the inset of Fig. 2(e). Since these waveguides share the same coupling and bending losses, by measuring their transmission as a function of the differential length, we can extract the propagation loss. Figure 2(e) shows the measurement results, where the propagation loss of straight waveguides for the $\text{TE}_{0,\text{tele}}$ mode is measured to be 0.54 dB/cm , a small value that represents the state-of-the-art quality of LN nanophotonic waveguides [35,44,45]. Together with the overall fiber-to-fiber transmission of a straight waveguide [for example, see Fig. 2(f)], we obtained a fiber-to-chip coupling loss of about 5 dB/facet .

To demonstrate SHG, we employed a straight waveguide with a length of about 8 mm. We launched a telecom-band continuous-wave (CW) laser into the device, with the setup shown in Fig. 2(a). By scanning the laser wavelength, we were able to measure the efficiency spectrum of SHG. One example is shown in Fig. 3(a), which shows a phase-matched pump wavelength of 1559 nm at a temperature of 18.7°C . The main lobe of the recorded efficiency spectrum agrees well with the theoretical

expectation from the function $\text{sinc}^2(\Delta L/2) = \left[\frac{\sin(\Delta L/2)}{\Delta L/2}\right]^2$. Interestingly, the efficiency spectrum exhibits multiple significant sidelobes, which are likely introduced by potential nonuniformity of the waveguide thickness along its total length, since the phase-matching condition of SHG is very sensitive to the waveguide geometry.

By fixing the pump wavelength at 1559.06 nm , where the peak conversion efficiency is located, we observed coherent radiation from its second harmonic at 779.53 nm , as shown clearly in Fig. 3(b). By varying the pump power, we recorded the SHG power that is plotted in Fig. 3(c). The second harmonic shows a quadratic power dependence on the pump that agrees very well with the theoretical expectation. Fitting the experimental data, we obtained an on-chip conversion efficiency of $4.7\% \text{ W}^{-1}$ [see Fig. 3(c)].

The recorded SHG efficiency is smaller than the theoretical value given by $\eta L^2 (= 14.2\% \text{ W}^{-1})$, primarily due to the nonzero propagation losses of the waveguide. The LN waveguide exhibits a propagation loss of 0.54 dB/cm at the pump wavelength [see Fig. 2(e)]. That at the second harmonic, however, is challenging to characterize with the current setup, due to the difficulty in effective light coupling into a high-order waveguide mode. As a rough estimate, we assume it to be $\sim 2.16 \text{ dB/cm}$, since the propagation loss is dominated by the Rayleigh scattering from

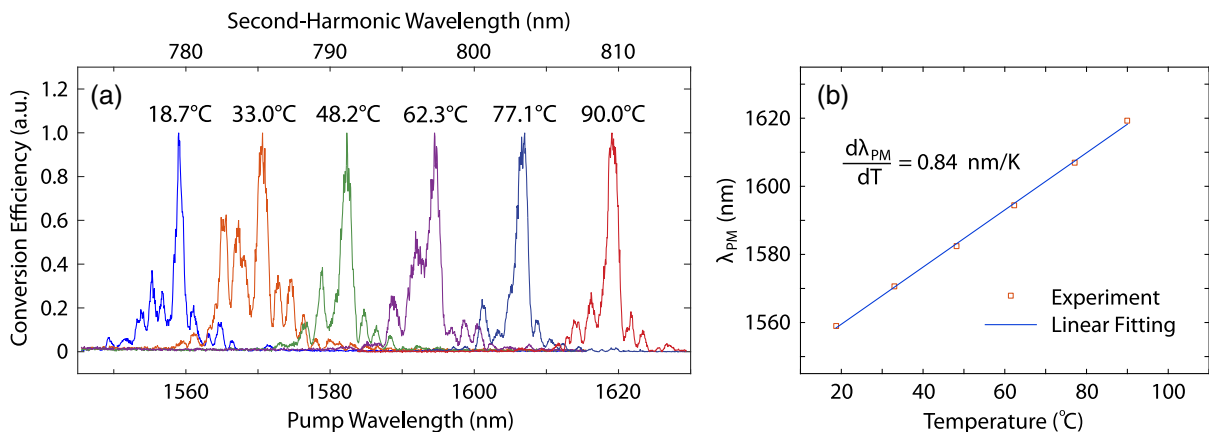


Fig. 4. Thermal tuning of SHG. (a) Conversion efficiency spectra at different temperatures. Each spectrum is normalized by its peak value for clear comparison. (b) Measured phase-matched pump wavelength λ_{PM} as a function of temperature.

the sidewall roughness that scales with the wavelength as $1/\lambda^2$ [46]. As a result, the theoretical conversion efficiency is estimated to be $\sim 9.1\% \text{ W}^{-1}$ for the 8-mm-long waveguide, after we take into account these propagation losses. On the other hand, the potential nonuniformity in the waveguide geometry might also impact the conversion efficiency to a certain extent. In addition, the coupling loss of the high-order second-harmonic light is likely underestimated due to its large mode mismatch with the focused fiber modes, leading to a conservative estimation of the recorded conversion efficiency. Therefore, the measured conversion efficiency can be improved in the future by reducing the waveguide sidewall scattering loss (say, with an oxide cladding), by increasing the uniformity in the wafer thickness with appropriate chemical mechanical polishing, and by optimizing coupling of the second-harmonic light (say, with a separately designed on-chip coupler [23]).

To show the spectral tuning capability of our device, we varied the device temperature from 18.7°C to 90.0°C and measured the SHG efficiency spectra. The recorded peak efficiencies are similar to that shown at 18.7°C [see Fig. 3(c)]. Figure 4(a) presents the peak-normalized spectra at different temperatures. It shows clearly that the SHG spectrum shifts towards longer wavelengths when the device temperature increases. By mapping the phase-matched pump wavelength as a function of temperature, we obtained Fig. 4(b), showing an experimentally measured tuning slope of $\frac{d\lambda_{\text{PM}}}{dT} = 0.84 \text{ nm/K}$, almost 1 order of magnitude larger than that achieved by type-0 SHG in LN [11,12,36]. The experimental results agree well with our simulations [see Fig. 1(f)]. A slightly larger experimental value of the tuning slope is likely due to positive contributions by pyroelectric [47] and thermal expansion effects [48] in the waveguide cross section, which were not taken into account in the simulations.

4. CONCLUSION

In conclusion, we have demonstrated highly tunable efficient SHG in an LN nanophotonic waveguide. The LN waveguide exhibits a high optical quality with a propagation loss as low as 0.54 dB/cm in the telecom band, which represents the state-of-the-art quality of LN nanophotonic waveguides reported to date [35,44,45]. In particular, we took advantage of the strong thermo-optic birefringence of LN to achieve thermal tuning of the SHG wavelength, with a tuning slope of 0.84 nm/K for a telecom-band pump, significantly higher than that offered by type-0 SHG in LN. At the same time, thanks to the tight mode confinement and a large spatial mode overlap, our waveguide exhibits a high theoretical normalized conversion efficiency of $22.2\% \text{ W}^{-1} \text{ cm}^{-2}$, even for the type-I intermodal phase matching, which is comparable to that of type-0 SHG in typical PPLN and LN nanophotonic waveguides utilizing the largest nonlinear term d_{33} . Our waveguide design enabled us to experimentally record an SHG efficiency of $4.7\% \text{ W}^{-1}$ inside a waveguide only 8 mm long.

We have demonstrated large tuning of type-I SHG by exploiting the strong thermo-optic birefringence of LN. In fact, this technique can also be applied to increase tunability of other nonlinear parametric processes in LN, such as DFG and SFG, by employing interband optical waves with different polarizations. Compared with type-0 phase matching widely studied in LN for a higher efficiency, type-I and type-II processes in LN nanophotonic waveguides offer a much larger thermal tunability while maintaining a high efficiency, showing great promise for tunable

and efficient on-chip nonlinear wavelength conversion that produces coherent radiation from visible to mid-infrared wavelengths.

APPENDIX A

1. Device Fabrication

Starting from a Z-cut LN-on-insulator wafer by NANOLN, we used electron-beam lithography with ZEP520A as the resist for device patterning, and argon ion milling for etching. Next, in order to remove the remaining resist and material residuals, we treated the chip with oxygen plasma followed by diluted hydrofluoric acid. Finally, we diced the chip and polished the facets for light coupling.

2. Experimental Setup

Pump light from a CW tunable telecom-band laser was coupled via a lensed fiber into the device chip, which was placed on top of a thermoelectric cooler that controls the temperature. At the waveguide output, pump light was collected together with the frequency-doubled light by a second lensed fiber. After being separated from its second harmonic by a 780/1550 wavelength division multiplexer, the telecom pump light was directed to an InGaAs detector for characterization, while the generated visible light was sent to a spectrometer for detection. A fiber polarization controller was used for optimal coupling from the input lensed fiber to the wanted waveguide mode, and variable optical attenuators were employed to study the power dependence of SHG. The spectrometer was cooled by liquid nitrogen for a high sensitivity.

3. SHG Spectrum Measurement

After aligning lensed fibers to the waveguide for optimal coupling, we scanned the telecom-band pump laser, with the spectrometer recording generated second-harmonic light during the whole laser scanning period. This process was repeated for different temperatures, which were controlled by the thermoelectric cooler under the device chip, to obtain the temperature dependence of the SHG spectrum.

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REFERENCES

1. A. L. Schawlow and C. H. Townes, "Infrared and optical masers," *Phys. Rev.* **112**, 1940–1949 (1958).

2. T. H. Maiman, "Stimulated optical radiation in ruby," *Nature* **187**, 493–494 (1960).
3. G. P. Agrawal, *Fiber-Optic Communication Systems* (Wiley, 2012).
4. J. Hodgkinson and R. P. Tatam, "Optical gas sensing: a review," *Meas. Sci. Technol.* **24**, 012004 (2012).
5. M.-G. Suh, Q.-F. Yang, K. Y. Yang, X. Yi, and K. J. Vahala, "Microresonator soliton dual-comb spectroscopy," *Science* **354**, 600–603 (2016).
6. T. Udem, R. Holzwarth, and T. W. Hänsch, "Optical frequency metrology," *Nature* **416**, 233–237 (2002).
7. F. Vollmer and S. Arnold, "Whispering-gallery-mode biosensing: label-free detection down to single molecules," *Nat. Methods* **5**, 591–596 (2008).
8. R. L. Byer, *Optical Parametric Oscillators* (Academic, 1975), Vol. 1.
9. M. H. Dunn and M. Ebrahimzadeh, "Parametric generation of tunable light from continuous-wave to femtosecond pulses," *Science* **286**, 1513–1517 (1999).
10. I. Breunig, D. Haertle, and K. Buse, "Continuous-wave optical parametric oscillators: recent developments and prospects," *Appl. Phys. B* **105**, 99–111 (2011).
11. M. M. Fejer, G. Magel, D. H. Jundt, and R. L. Byer, "Quasi-phase-matched second harmonic generation: tuning and tolerances," *IEEE J. Quantum Electron.* **28**, 2631–2654 (1992).
12. L. E. Myers, R. Eckardt, M. Fejer, R. Byer, W. Bosenberg, and J. Pierce, "Quasi-phase-matched optical parametric oscillators in bulk periodically poled LiNbO₃," *J. Opt. Soc. Am. B* **12**, 2102–2116 (1995).
13. J.-P. Meyn and M. Fejer, "Tunable ultraviolet radiation by second-harmonic generation in periodically poled lithium tantalate," *Opt. Lett.* **22**, 1214–1216 (1997).
14. L. Moretti, M. Iodice, F. G. D. Corte, and I. Rendina, "Temperature dependence of the thermo-optic coefficient of lithium niobate, from 300 to 515 K in the visible and infrared regions," *J. Appl. Phys.* **98**, 036101 (2005).
15. R. Luo, H. Jiang, H. Liang, Y. Chen, and Q. Lin, "Self-referenced temperature sensing with a lithium niobate microdisk resonator," *Opt. Lett.* **42**, 1281–1284 (2017).
16. W. Weng, P. S. Light, and A. N. Luiten, "Ultra-sensitive lithium niobate thermometer based on a dual-resonant whispering-gallery-mode cavity," *Opt. Lett.* **43**, 1415–1418 (2018).
17. W. J. Tropf, M. E. Thomas, and T. J. Harris, "Properties of crystals and glasses," in *Handbook of Optics*, M. Bass, E. W. van Stryland, and D. R. Williams, eds., 2nd ed. (McGraw-Hill, 1995–2001), Vol. 2, Chap. 33, pp. 33.1–33.101.
18. N. E. Yu, J. H. Ro, M. Cha, S. Kurimura, and T. Taira, "Broadband quasi-phase-matched second-harmonic generation in MgO-doped periodically poled LiNbO₃ at the communications band," *Opt. Lett.* **27**, 1046–1048 (2002).
19. L. Scaccabarozzi, M. Fejer, Y. Huo, S. Fan, X. Yu, and J. S. Harris, "Enhanced second-harmonic generation in AlGaAs/AlxOy tightly confining waveguides and resonant cavities," *Opt. Lett.* **31**, 3626–3628 (2006).
20. K. Rivoire, Z. Lin, F. Hatami, W. T. Masselink, and J. Vučković, "Second harmonic generation in gallium phosphide photonic crystal nanocavities with ultralow continuous wave pump power," *Opt. Express* **17**, 22609–22615 (2009).
21. J. S. Levy, M. A. Foster, A. L. Gaeta, and M. Lipson, "Harmonic generation in silicon nitride ring resonators," *Opt. Express* **19**, 11415–11421 (2011).
22. D. Duchesne, K. A. Rutkowska, M. Volatier, F. Légaré, S. Delprat, M. Chaker, D. Modotto, A. Locatelli, C. D. Angelis, M. Sorel, D. N. Christodoulides, G. Salamo, R. Arès, V. Aimez, and R. Morandotti, "Second harmonic generation in AlGaAs photonic wires using low power continuous wave light," *Opt. Express* **19**, 12408–12417 (2011).
23. X. Guo, C.-L. Zou, and H. X. Tang, "Second-harmonic generation in aluminum nitride microrings with 2500%/W conversion efficiency," *Optica* **3**, 1126–1131 (2016).
24. E. Timurdogan, C. V. Poulton, M. Byrd, and M. Watts, "Electric field-induced second-order nonlinear optical effects in silicon waveguides," *Nat. Photonics* **11**, 200–206 (2017).
25. A. Billat, D. Grassani, M. H. Pfeiffer, S. Kharitonov, T. J. Kippenberg, and C.-S. Brès, "Large second harmonic generation enhancement in Si₃N₄ waveguides by all-optically induced quasi-phase-matching," *Nat. Commun.* **8**, 1016 (2017).
26. H. Hu, R. Ricken, and W. Sohler, "Lithium niobate photonic wires," *Opt. Express* **17**, 24261–24268 (2009).
27. R. Geiss, S. Saravi, A. Sergeyev, S. Diziain, F. Setzpfandt, F. Schreppe, R. Grange, E.-B. Kley, A. Tünnermann, and T. Pertsch, "Fabrication of nanoscale lithium niobate waveguides for second-harmonic generation," *Opt. Lett.* **40**, 2715–2718 (2015).
28. J. Lin, Y. Xu, Z. Fang, M. Wang, J. Song, N. Wang, L. Qiao, W. Fang, and Y. Cheng, "Fabrication of high-q lithium niobate microresonators using femtosecond laser micromachining," *Sci. Rep.* **5**, 8072 (2015).
29. L. Chang, Y. Li, N. Volet, L. Wang, J. Peters, and J. E. Bowers, "Thin film wavelength converters for photonic integrated circuits," *Optica* **3**, 531–535 (2016).
30. J. Lin, Y. Xu, J. Ni, M. Wang, Z. Fang, L. Qiao, W. Fang, and Y. Cheng, "Phase-matched second-harmonic generation in an on-chip LiNbO₃ microresonator," *Phys. Rev. Appl.* **6**, 014002 (2016).
31. C. Wang, X. Xiong, N. Andrade, V. Venkataraman, X.-F. Ren, G.-C. Guo, and M. Lončar, "Second harmonic generation in nano-structured thin-film lithium niobate waveguides," *Opt. Express* **25**, 6963–6973 (2017).
32. A. Rao, J. Chiles, S. Khan, S. Toroghi, M. Malinowski, G. F. Camacho-González, and S. Fathpour, "Second-harmonic generation in single-mode integrated waveguides based on mode-shape modulation," *Appl. Phys. Lett.* **110**, 111109 (2017).
33. R. Luo, H. Jiang, S. Rogers, H. Liang, Y. He, and Q. Lin, "On-chip second-harmonic generation and broadband parametric down-conversion in a lithium niobate microresonator," *Opt. Express* **25**, 24531–24539 (2017).
34. C. Wang, Z. Li, M.-H. Kim, X. Xiong, X.-F. Ren, G.-C. Guo, N. Yu, and M. Lončar, "Metasurface-assisted phase-matching-free second harmonic generation in lithium niobate waveguides," *Nat. Commun.* **8**, 2098 (2017).
35. R. Wolf, I. Breunig, H. Zappe, and K. Buse, "Cascaded second-order optical nonlinearities in on-chip micro rings," *Opt. Express* **25**, 29927–29933 (2017).
36. L. Gui, "Periodically poled ridge waveguides and photonic wires in LiNbO₃ for efficient nonlinear interactions," Ph.D. dissertation (University of Paderborn, 2010).
37. X.-C. Yu, B.-B. Li, P. Wang, L. Tong, X.-F. Jiang, Y. Li, Q. Gong, and Y.-F. Xiao, "Single nanoparticle detection and sizing using a nanofiber pair in an aqueous environment," *Adv. Mater.* **26**, 7462–7467 (2014).
38. C. C. Evans, C. Liu, and J. Suntivich, "TiO₂ nanophotonic sensors for efficient integrated evanescent Raman spectroscopy," *ACS Photon.* **3**, 1662–1669 (2016).
39. W. Yu, W. C. Jiang, Q. Lin, and T. Lu, "Cavity optomechanical spring sensing of single molecules," *Nat. Commun.* **7**, 12311 (2016).
40. D. E. Zelmon, D. L. Small, and D. Jundt, "Infrared corrected Sellmeier coefficients for congruently grown lithium niobate and 5 mol.% magnesium oxide-doped lithium niobate," *J. Opt. Soc. Am. B* **14**, 3319–3322 (1997).
41. R. Boyd, *Nonlinear Optics*, 3rd ed. (Academic, 2008).
42. K. R. Parameswaran, J. R. Kurz, R. V. Roussev, and M. M. Fejer, "Observation of 99% pump depletion in single-pass second-harmonic generation in a periodically poled lithium niobate waveguide," *Opt. Lett.* **27**, 43–45 (2002).
43. D. A. Roberts, "Simplified characterization of uniaxial and biaxial nonlinear optical crystals: a plea for standardization of nomenclature and conventions," *IEEE J. Quantum Electron.* **28**, 2057–2074 (1992).
44. M. Zhang, C. Wang, R. Cheng, A. Shams-Ansari, and M. Lončar, "Monolithic ultra-high-Q lithium niobate microring resonator," *Optica* **4**, 1536–1537 (2017).
45. I. Krasnokutskaya, J.-L. J. Tambasco, X. Li, and A. Peruzzo, "Ultra-low loss photonic circuits in lithium niobate on insulator," *Opt. Express* **26**, 897–904 (2018).
46. P. K. Tien, "Light waves in thin films and integrated optics," *Appl. Opt.* **10**, 2395–2413 (1971).
47. H. Lu, B. Sadani, G. Ulliac, C. Guyot, N. Courjal, M. Collet, F. I. Baida, and M.-P. Bernal, "Integrated temperature sensor based on an enhanced pyroelectric photonic crystal," *Opt. Express* **21**, 16311–16318 (2013).
48. Y. Kim and R. Smith, "Thermal expansion of lithium tantalate and lithium niobate single crystals," *J. Appl. Phys.* **40**, 4637–4641 (1969).